TECHNICAL BRIEF · RECOGNITION AND MEASUREMENT

NexGen iMATCH

The Recognition Engine, Its Evaluation, and What the Measurements Establish

Field	Detail
Deployed recogniser	ArcFace `w600k_r50` (ResNet-50, WebFace600K), 512-dimension L2-normalised embedding
Approved successor	CVLface ViT-B + KP-RPE + AdaFace (WebFace12M), with thresholds re-derived
Detector / aligner	SCRFD (`det_10g`) · 5-point similarity transform to 112×112 · DFA aligner for non-pack crops
Operating thresholds	match 0.2871 · review 0.2153 · verify 0.2871 (derived at FMR = 0.1%)
Evaluation corpora	LFW, CFP-FF, CFP-FP, AgeDB-30, CALFW, CPLFW, IJB-B, IJB-C, TinyFace, QMUL-SurvFace, SCface
Compute	One NVIDIA RTX A3000 Laptop GPU, 6 GB — the binding constraint on every design choice
Reporting rule	Accuracy is never blended across operating conditions. Clean and surveillance performance are reported separately.

The one claim this document makes, and the one it does not

Makes: every figure here was produced under a named protocol, on a stated corpus, by a harness validated against published external results, and each figure names the artifact file that produced it.
Does not make: that the recognition model is our own contribution. Clean-imagery accuracy is at the level of the open-source state of the art *because it is the open-source state of the art*. What is ours is the measurement framework, the protocol discipline, and the results that framework produced.

Contents

1. Purpose and scope

This brief documents the recognition engine of NexGen iMATCH: the models used, the protocol under which they were evaluated, the results by operating condition, the interventions that were tested and rejected, and the limits the measurements establish. It is written for an engineer or scientist who intends to check it.

Three conventions are used throughout, and they are the reason this document can be checked rather than believed. **First**, no accuracy figure appears without its corpus and its protocol. **Second**, where a figure has been superseded, the withdrawn value, its replacement and the defect that caused it are all recorded. **Third**, where a decision rule was registered before an experiment, the result is reported against that rule even when the rule was not met.

2. Pipeline, layer by layer

Layer	Implementation	Notes that matter
1. Detection	SCRFD (`det_10g`, InsightFace buffalo_l pack) on ONNX Runtime, detection size 640×640	Padded retry for small faces. GPU binding is asserted by inspecting post-construction session providers, so silent CPU fallback cannot occur unnoticed.
2. Alignment	Five-point similarity transform (Umeyama) to a 112×112 crop; DFA aligner on the ViT path	Landmarks also yield head pose. On non-pack crops the DFA aligner is mandatory and is worth approximately 51 points on TinyFace — see §9.1.
3. Quality	Eight-component quality report with blocking and advisory reasons separated; plus blur σ, noise σ and compression-quality estimation	Every degradation estimate carries a confidence, and every downstream trigger is gated on it. Sharpness uses Laplacian variance; effective spectral cut-off gives true, as opposed to stored, resolution.
4. Enhancement	Track A: six classical operators, no weights. Track B: Real-ESRGAN, NAFNet, CodeFormer, GFPGAN	A pre-processing layer in front of an unmodified recogniser. Track B is off by default; weights are checksum-pinned. Evaluated and rejected for recognition — §9.4.
5. Embedding	ArcFace `w600k_r50` → 512-d, L2-normalised. Alternatives: ViT-B KP-RPE AdaFace, and that model with a rank-8 LoRA adapter	Test-time augmentation is horizontal flip only, because ArcFace is trained with flip augmentation.
6. Gallery	Per-tenant in-memory matrix, rebuilt from the database on first use; templates encrypted at rest	The tenant identifier is a required argument to search, so cross-tenant comparison has no code path.
7. Matching	Cosine similarity, which on L2-normalised vectors reduces to a dot product. Exhaustive matmul — exact, not approximate	An approximate index was measured and deferred: below roughly 10,000 templates there is nothing to win — §9.5.
8. Decision	Single-source threshold configuration; three bands — match, review, no match	Quality, liveness and morphing signals contribute an advisory score; they never override the decision. Adjudication is performed by a human.

3. Models: deployed and evaluated

3.1 What is deployed, and why

Model	Architecture / training set	Status
`w600k_r50`	ResNet-50, WebFace600K (InsightFace buffalo_l)	**Deployed**, and retained permanently as the reference baseline against which every claim is measured
`glintr100`	ResNet-100, Glint360K (antelopev2)	Evaluated. Higher on clean packs, materially worse on degraded imagery — not adopted
`vit_kprpe_wf12m`	ViT-B + KP-RPE + AdaFace, WebFace12M (115M parameters)	**Approved to replace** the R50, with thresholds re-derived (§4.3)
`vit_kprpe_lora_r8`	The above, plus a rank-8 LoRA adapter trained on real low-resolution imagery	Gate result: two criteria passed, one failed — §10

The model-selection decision that is easiest to get wrong

On clean benchmark packs `glintr100` scores higher than the deployed R50 (97.35 against 97.16 mean). On TinyFace the R50 is decisively better: **33.13% against 17.37%** true-accept at a false-accept rate of 0.1%. Choosing the model by its clean-benchmark ranking would have halved performance in the operating condition this system exists for. **The deployment is therefore a deliberate trade, not an oversight**, and it is recorded as one.

3.2 What is not claimed

The weights are pretrained and publicly available. No original recognition architecture is claimed, and clean-imagery parity with published results is not presented as an achievement — it is the expected consequence of using the same weights under a correct protocol. The value of that parity is diagnostic: it is what establishes that the measurement harness is not producing artefacts.

4. Threshold calibration

4.1 The rule

The operating threshold is derived at a **false-match rate of 0.1%** on AgeDB-30, using 32,000 impostor pairs matched *within* subgroup — same gender, same age band. Cross-group impostor pairs are trivially easy and would deflate every false-match figure. One global threshold is used; per-group thresholds would conceal the differential that §11 exists to expose.

4.2 The decision record

The threshold was raised from 0.20 to 0.2871 after a real comparison of two different people returned a similarity of 0.2405 and was reported as supporting the same-person proposition. This was not a defect: the pair fell between the deployed threshold and the target false-match point, and it is a direct instance of the 1.19% false-match rate already measured at 0.20. The prior value had been chosen to maximise *accuracy*, which weights a miss and a false match equally — a weighting that is wrong for forensic use.

Group	FNMR @ 0.20	FNMR @ 0.2871	FMR @ 0.20	FMR @ 0.2871
All	3.30%	**6.32%**	**1.19%**	**0.10%**
Female	4.88%	8.45%	1.70%	0.16%

Group	FNMR @ 0.20	FNMR @ 0.2871	FMR @ 0.20	FMR @ 0.2871
Male	2.21%	4.86%	0.68%	0.03%
Age 0–25	7.61%	**14.75%**	1.21%	0.10%
Age 26–40	3.24%	5.78%	1.26%	0.05%
Age 41–55	2.14%	3.91%	1.10%	0.15%
Age 56+	2.90%	6.12%	1.19%	0.09%

40,098 AgeDB pairs — 8,098 genuine, 32,000 impostor. False matches fell approximately twelvefold (about 1 in 84 to about 1 in 1,000); missed genuine pairs roughly doubled. Both the benefit and the cost are recorded.

4.3 Thresholds do not carry across models

0.2871 is the FMR = 0.1% operating point *for the R50*. Re-derived by the identical rule for the ViT: match **0.2371**, review **0.1778**. For the LoRA-adapted model: match **0.2561**, review **0.1921**. Carrying a threshold across a model change would relocate the operating point to a different place on the ROC curve while continuing to look calibrated. This is treated as a blocking requirement of any model swap, not a follow-up task.

5. Evaluation methodology

5.1 Protocol discipline

- **Ten-fold cross-validation** on the verification packs, with the threshold fitted on nine folds and applied to the held-out fold. No threshold is ever fitted on the data it is scored against.
- **Official protocols only** where one exists — official pair lists, official gallery and probe splits, official template pooling. Comparison sets constructed by convenience were the cause of two withdrawn figures (§13).
- **Reference populations are identity-disjoint by construction**, and the disjointness is asserted in tests rather than assumed.
- **Impostor pools are audited** before use. A pool whose audit fails is refused rather than reported — for example, capacity figures on unlabelled protocol pools were withdrawn entirely because saturation and contamination could not be excluded.
- **Decision rules are registered in code before the run**, with explicit pass and fail thresholds and a confidence-interval requirement.

5.2 Harness validation against an external protocol

The most important check in this document is not an accuracy figure — it is the demonstration that the harness reproduces published results on a protocol we did not design.

	IJB-B	IJB-C
Faces	227,630	469,375
Templates	12,115	23,124
Pairs (genuine / impostor)	10,270 / 8,000,000	19,557 / 15,638,932
TAR @ FAR = 1e-3	97.03%	98.13%
TAR @ FAR = 1e-4	**95.44%**	**96.91%**
TAR @ FAR = 1e-5	91.07%	94.43%
TAR @ FAR = 1e-6	43.13% *[CI 26.73–49.80]*	86.73% *[CI 82.65–88.96]*
Impostor pairs supporting 1e-6	**8**	**16**

Why the 1e-6 row is recorded but not published

Both figures rest on a threshold set by single-digit or low-double-digit impostor pairs. A Poissonised bootstrap of the impostor tail gives confidence intervals 6.3 points wide on IJB-C and 23.1 points wide on IJB-B. An operating point needs roughly 400 exceedances for ±10% precision — that is $4{\times}10^8$ impostor pairs, and the official IJB-C protocol supplies $1.56{\times}10^7$. The table therefore carries the supporting pair count beside the number so the two cannot be read apart. **Neither figure may be quoted as a performance result.**

A related estimator change is recorded for the same reason: thresholds are now taken as the exact k-th order statistic rather than by interpolating between adjacent impostor scores. At FAR ≥ 1e-5 the two agree within 0.02 points; at 1e-6 interpolation was manufacturing precision the data does not contain. Superseded: IJB-C 86.83% → 86.73%, IJB-B 43.43% → 43.13%.

Analysis of the IJB-B tail. Eight impostor pairs sit approximately 0.35 cosine above the rest of the impostor distribution and above the genuine median, so the 1e-6 threshold lands above half of all true matches and true-accept collapses by 48 points between two adjacent decades. Both backbones meet the same eight pairs, which is why neither can move the figure, and no impostor pair exceeds cosine 0.99, so they are not duplicate images. **That cell measures eight pathological or mislabelled templates, not recognition.**

Caveat carried deliberately. The published-ArcFace comparison band (~94–95% on IJB-B and ~96–97% on IJB-C at FAR = 1e-4) is recalled from the literature and is not yet cited to a verified source in this repository. It is the sole external anchor for the harness-validation claim, and it is flagged as requiring pinning to a specific paper and table before the claim is relied upon outside the team.

6. Results on clean imagery

Pack	R50 (deployed)	ViT-B KP-RPE	ViT + LoRA r8
LFW	99.78	99.75	**99.85**
CFP-FF	99.87	99.86	99.84
CFP-FP	97.44	97.33	**97.56**
AgeDB-30	98.15	97.90	97.77
CALFW	95.95	95.82	**96.05**
CPLFW	94.47	94.40	94.47

Ten-fold accuracy, percent. The ViT is fractionally behind the R50 on every clean pack — all within fold standard deviation, but consistently so, and it is recorded as a real if small cost rather than rounded to “equal”. The adoption argument is that these losses of at most 0.25 points fall on saturated benchmarks that carry no operational information, against a threefold to fivefold gain in the condition the system exists for.

One caveat on the CFP-FP column. The `.bin` protocol packs differ between the training bundles that ship them; three distinct CFP-FP variants exist. The figures above were measured on the outlier pack and understate all configurations by roughly 1.5 points. Rankings are unaffected because the pack is common to every configuration, but the absolute values require re-running before external use.

7. Results on surveillance imagery

Two corpora, complementary by structure and neither a substitute for the other: TinyFace has a 153,000-image distractor gallery but **zero unmated probes**; QMUL-SurvFace has true unmated probes but a small gallery.

7.1 TinyFace — official protocol, 153,428-image reference population

Metric	R50	ViT-B KP-RPE	Ratio
TAR @ FAR = 1%	35.55%	**70.69%**	2.0×
TAR @ FAR = 0.1%	20.59%	**60.77%**	3.0×
TAR @ FAR = 0.01%	12.07%	**48.62%**	4.0×
TAR @ FAR = 1e-5	7.38%	**35.03%**	4.7×
Rank-1 (155,997 gallery)	32.93%	**68.20%**	2.1×
Identity bits, median	4.73	**12.95**	+8.2 bits
Supportable gallery at 50% rank-1	13	**3,965**	305×
Cllr	0.6662	**0.4920**	

The ratio widens as the false-accept rate tightens — 2.0× at 1%, 4.7× at 1e-5 — which is the signature of genuinely better non-mate separation rather than a threshold shift.

7.2 QMUL-SurvFace — official open-set protocol

Metric	R50	ViT-B KP-RPE	ViT + LoRA
TAR @ FAR = 1%	8.68%	17.06%	**38.66%**
TAR @ FAR = 0.1%	2.31%	4.98%	**22.35%**

Metric	R50	ViT-B KP-RPE	ViT + LoRA
TAR @ FAR = 1e-5	0.11%	0.34%	**2.91%**
Rank-1	2.68%	6.06%	**13.06%**
Identity bits, median	2.21	2.92	**4.79**
TPIR @ 1% FPIR	**0.00%**	**0.00%**	**1.38%**
TPIR @ 10% FPIR	0.17%	0.78%	**5.97%**
Cllr	0.9047	0.8400	**0.7115**

8. The capacity framework

This is the part of the programme that is genuinely ours, and it is the reason the results above are interpretable rather than merely collected.

8.1 The question it answers

How much identity information does one observation actually carry, and how much does a gallery of a given size demand? Expressed in bits, the two can be compared directly — and the comparison predicts whether a given deployment is feasible *before* it is attempted.

Corpus	Bits delivered (median)	Bits a gallery demands	Consequence
TinyFace	4.73 (R50) → 12.95 (ViT)	— (no unmated probes; open-set unmeasurable here)	Ranking improves sharply with a better backbone
QMUL-SurvFace	2.21 (R50) → 2.92 (ViT) → 4.79 (LoRA)	≈ 11.5 for its 2,965-entry gallery	Short by a factor of roughly $2^{8.6}$; thresholded open-set identification stays near zero

8.2 The failure mechanism, measured directly

Backbone	Mean top-1 score, enrolled probe	Mean top-1 score, stranger	Gap
`w600k_r50`	+0.5703	+0.5705	−**0.0002**
`vit_kprpe_wf12m`	+0.4782	+0.4771	**+0.0011**

The maximum-over-gallery score carries essentially no information about whether the probe's identity is enrolled at all. With 2,965 gallery entries and 25-pixel faces there is always somebody who resembles a stranger as closely as the true mate resembles its own probe. **No threshold can separate distributions whose means differ by 0.001.**

The result that gives the framework its standing

The capacity analysis stated *before* the backbone swap that a better model would improve ranking while leaving thresholded open-set identification near zero. The swap improved verification two- to threefold and rank-1 by 2.3× — and moved true-positive identification at 1% false-positive identification rate by **exactly nothing**, from 0.00% to 0.00%. A framework that predicts a null result correctly, against the intuition that a better model must help, is a predictive instrument rather than a descriptive statistic.

8.3 Calibration is exhausted; discrimination is the constraint

The log-likelihood-ratio cost decomposes into a discrimination term and a calibration term. On both surveillance corpora the recoverable calibration component is at most 0.6% of the total loss. That is a quantitative statement that **no amount of further work on the evidence layer, score normalisation or reporting can improve accuracy** — only more identity information can.

8.4 A caveat on the bit estimates

On TinyFace, censoring rose from 2.0% to 13.1% after the backbone swap — 13% of genuine pairs now out-score all 50 million sampled impostors — so the ViT's true capacity is *higher* than the 12.95-bit median states, and the estimator is at its ceiling. A larger impostor pool would be required to resolve it. On QMUL censoring is 0.000%, so those figures are fully resolved.

9. Interventions evaluated, with verdicts

Each of the following was tested against a rule fixed beforehand. Four were adopted, five rejected, one deferred on measurement. The rejections are reported at the same length as the adoptions.

Intervention	Verdict	Evidence
ViT-B KP-RPE AdaFace backbone	**ADOPT**	3.0× TAR at FAR = 0.1% on TinyFace, 2.1× rank-1, with a recorded cost of 0.03–0.25 points on clean packs
DFA aligner for non-pack crops	**ADOPT**, mandatory	Worth approximately +51 points on TinyFace — §9.1
Score normalisation (margin, per-probe z-norm)	**ADOPT**	27× TPIR at 1% FPIR, at zero training cost. Justified by a prior measurement: per-probe impostor means vary with standard deviation 0.0573 across probes — 57× larger than the 0.001 signal, so probe-dependent hubness was drowning the evidence
Embedding centering + WCCN	**ADOPT**, operator-selectable	41× TPIR at 1% FPIR, fitted on the gallery only. Costs about one point of rank-1, so it is exposed as a mode rather than hard-coded
Multi-frame template averaging	**REJECT**	Measured counterproductive — §9.2
Forward-operator / degradation-aware modelling	**REJECT** for camera-to-camera; narrow licence for the asymmetric case	§9.3
Image enhancement as a recognition aid	**REJECT**	§9.4
Multi-model ensembling	**REJECT**	The three-model ensemble lost to the best single model on four of five datasets at three times the compute
Approximate nearest-neighbour search	**DEFER** on measurement	§9.5
Probabilistic identity embeddings	**DEFER**	Uncertainty is already handled in the evidence layer; moving it into the embedding re-expresses the deficit rather than reducing it, and adds no bits

9.1 Two silent integration defects in the ViT path

Defect	Observed effect	Why the obvious gate missed it
KP-RPE keypoints are [0,1]-normalised, not pixel units	LFW 56.70% against an expected 99.75% — near chance, and silent	—
Canonical keypoints supplied to KP-RPE on crops that are not ArcFace-aligned	TinyFace 15.19% — **below the R50 incumbent**. Withdrawn and archived	Protocol pack crops *are* aligned, so the LFW gate passed at 99.75% while the assumption was false for every surveillance corpus, whose crops are detector crops with margin

The generalisable lesson: a gate that validates tensor plumbing does not validate a semantic assumption. Both are now contract-tested, including an assertion that alignment remains the *default* path — if it ever becomes opt-in, degraded-imagery figures regress by roughly 50 points while still looking plausible.

9.2 Multi-frame averaging — rejected on measurement

Frames averaged	1	2	4	8	16
Open-set AUC	**0.5399**	0.4957	0.4821	0.4799	0.4785
Mated – stranger score gap	**+0.0167**	–0.0009	–0.0070	–0.0081	–0.0090

More frames makes stranger rejection strictly worse and drives AUC below chance. The mechanism follows from an earlier measurement: corpus-level condition leakage is +0.1088, far larger than the camera-level term of +0.0039. Every embedding carries a large shared acquisition-regime component. Averaging is a low-pass filter: it suppresses the idiosyncratic identity signal, which varies frame to frame, while preserving the shared condition component, which does not. This rejects *naive mean fusion*, not multi-frame evidence in principle — fusion can only help after the common-mode component is removed.

9.3 The forward-operator hypothesis — a pre-registered go/no-go

Question: does estimating the probe's imaging operator and applying it forward to the gallery beat simple resolution matching? **Rule, registered in code before any run:** pass at ≥ +2.0 points TAR at FAR = 0.1% with a 95% confidence interval excluding zero; fail at ≤ +0.5 or a confidence interval spanning zero.

Dataset	Character	B2r – B1 (points) [CI]	Verdict
TinyFace	surveillance to surveillance	+0.13 [–0.41, +1.84]	**FAIL**
QMUL-SurvFace	surveillance to surveillance	+0.13 [–2.20, +1.53]	**FAIL**
LFW (synthetic degradation)	high-resolution gallery, degraded probe	+3.55 [+2.02, +5.02]	**PASS**
SCface D1 (4.2 m)	real mugshot vs real CCTV	+0.11 [–0.43, +0.75]	**FAIL**
SCface D2 (2.6 m)	real mugshot vs real CCTV	+2.64 [0.00, +4.50]	**FAIL** — interval touches zero

Reading the verdict. Where probe and gallery share an imaging chain, the correct forward model reduces to a no-op — the method is coherent but there is no operator asymmetry to exploit. The single pass is on synthetically degraded imagery, where the operator being estimated is the one we applied, so part of that margin may be self-fulfilment. SCface was acquired specifically to settle that question on real optics, and it did not sustain the pass. An intermediate defect is worth recording: the first formulation applied the probe's *absolute* operator and produced double degradation at –8.0 points; the residual formulation beats it on every dataset, which is the independent confirmation that the diagnosis was correct.

9.4 Enhancement as a recognition aid — rejected

Six probe-side arms were registered against the existing loaders, embedder, metrics and paired bootstrap, with an adoption rule fixed before any GPU run: adopt at ≥ +2.0 points TAR at FAR = 0.1% over the untransformed baseline with a confidence interval excluding zero. Arm E0 is a plain-interpolation control, present specifically to separate “the recogniser prefers larger inputs” from genuine restoration.

Arm	E0 (control)	E1	E2	E3	E4	E5
vs baseline (points)	–0.22	–0.22	0.00	–0.33	–0.11	–0.44
Beats the interpolation control	—	No	No	No	No	No
Verdict	REJECT	REJECT	REJECT	REJECT	REJECT	REJECT

SCface D1, 10,010 pairs. **Every arm was rejected, and none beat the interpolation control.** The operational conclusion is stated plainly: restoration is optimised for human perceptual quality, which is a different objective from identity preservation. A face that looks sharper and matches no better is the expected outcome, not a surprise — and this negative result was written up as a deliverable, as the decision rule required.

9.5 Approximate search — deferred, with the measurement that defers it

Gallery size	1,000	10,000	100,000
Exhaustive search	0.207 ms	1.087 ms	15.98 ms
As a share of the ~14.7 ms probe encode	1.4%	7.4%	109%

Below roughly 10,000 templates there is nothing to win. Note also that the guarded branch in the code builds an *exact* inner-product index, so enabling it would buy a constant-factor speedup, not a change in complexity.

Real scaling requires an approximate index and the recall loss that comes with it — and for a forensic system that loss must be measured and accepted explicitly, because a missed candidate from an approximate index is an investigative lead that silently never surfaced.

10. LoRA fine-tuning on real low-resolution data

The prior fine-tuning attempt used *synthetically* degraded imagery and scored worse on every benchmark, worst on TinyFace. The diagnosis recorded at the time was that synthetic degradation does not match real low-resolution capture. This experiment is the direct test of that diagnosis.

Parameter	Value
Method	Rank-8 LoRA on 96 attention and MLP projections — 1.38M trainable of 115.1M (1.20%)
Objective	AdaFace margin head over 7,889 identities
Data	TinyFace and QMUL training splits, capped at 24 images per QMUL identity — 63,813 images, 7,889 identities
Contamination audit	Zero raw label overlap with either test split, verified before training
Alignment	Identical to inference (DFA aligner, pre-cached)
Cost	4 epochs, 53.9 minutes on the 6 GB A3000

10.1 The gate, registered before the run

Criterion	Target	Measured	Result
QMUL identity bits (median)	≥ 4.0	**4.79**	**PASS**
Clean-pack regression vs R50	≤ 0.5 points	–**0.38** (worst: AgeDB-30)	**PASS**
Condition leakage vs baseline	not above +0.0039	**+0.0051**	**FAIL**

On the failed criterion — recorded as a failure, not reinterpreted

Raw leakage rose from +0.0039 to +0.0051. By the rule registered in advance, that is a fail, and it is recorded as one. The honest complication, stated as a limitation of *the criterion* rather than as a rescue: absolute leakage is not scale-invariant, and the fine-tune compressed the whole impostor distribution (mean cross-identity similarity 0.193 → 0.040). Relative to that base, leakage is worse on that reading too. The decision-relevant form — false-accept amplification at a fixed operating point — is nearly unchanged at 1.3× → 1.4×. So the model is not obviously learning the camera in a way that changes decisions, but it is not cleaner either. **The correct response is a second experiment, not a redefinition:** the adversarial leakage term was specified in the implementation plan and was not trained in this run, and adding it is the direct test of whether the leakage cost is removable while keeping the gain in bits.

10.2 What it establishes

- **The discrimination deficit is partly addressable on this hardware.** A 1.2%-parameter adapter, 54 minutes and 63,813 real low-resolution images raised QMUL identity bits by 64% with no clean-set cost — the fine-tune in fact recovered accuracy the base ViT had lost, beating it on four of six clean packs.
- **The open-set metric moved for the first time.** True-positive identification at 1% false-positive identification rate was 0.00% for the R50 *and* for the base ViT; real low-resolution training data moved it to 1.38%, and the 10% figure by 7.7×. No other intervention of any kind in this programme has changed that metric.
- **1.38% is a starting point, not a capability.** The 11.5-bit gallery requirement is unchanged and 4.79 bits does not meet it.
- **The prior failure is explained.** Synthetic degradation lost on every benchmark; real low-resolution capture gained on nearly all. That was the documented hypothesis, and it is now confirmed.

11. Demographic differentials

Measured on AgeDB, whose filenames carry age and gender, using impostor pairs matched within subgroup. At the deployed threshold: women are falsely rejected approximately 1.7× as often as men (8.45% against 4.86%) and falsely matched approximately 5× as often (0.16% against 0.03%); the under-25 band is falsely rejected approximately 3.8× as often as the 41–55 band (14.75% against 3.91%).

Raising the threshold relocated the errors; it did not remove the gap. Both error types run against the same groups, which indicates a genuine accuracy differential rather than a threshold-placement artefact. A single aggregate figure conceals both: “6.32% false-non-match” looks acceptable while hiding a subgroup at 14.75%. Any deploying organisation must re-measure this on its own population.

12. Performance engineering

Measurement	Result
Encode one face (p50)	14.72 ms
Full 1:1 verification	31.04 ms
Gallery search	0.207 ms at 1k · 1.087 ms at 10k · 15.98 ms at 100k
Threading	Saturates at 4 workers — 1.86×, 131.9 requests/s. Eight workers gain nothing and degrade p99 by 2.3× to 100 ms
Request batching	2.82× — 551 requests/s, without the latency cost

If throughput becomes a constraint, the measured answer is a request-collecting queue, not more workers. These are engine-level figures; the full HTTP stack is not yet measured.

13. Defects found in our own measurement code

Twelve were found during the current campaign, four of which had produced published figures. All were found by our own validation, all were fixed, and all are recorded with symptom, cause and fix. This section exists because the most likely source of a wrong conclusion in a programme like this is not a wrong model — it is a measurement that was never correct.

#	Defect	Observed effect	Fix
1	Compression-quality estimation by recompression error	Returned quality 95 for a true quality of 35 at every image size tested	DCT-lattice fitting — 24 of 24 exact across six quality levels and four sizes
2	Confidence defined as best against second-best	Near-zero margin even when exactly right	Best against median
3	Channel averaging before compression estimation	Superimposed three quantisation lattices; 76.0% accuracy	Green-channel estimation; 100.0%
4	Trustworthiness flag from blur fit alone	Flag true while the compression estimate was wrong by 60 points	Conjunction with a per-component breakdown
5	Operator unit order in the degradation arm	Probe-space sigma applied before decimation — one eighth of the intended operator	Decimate first, apply in probe space
6	Absolute rather than residual operator	Double degradation, –8.0 points	Residual composition
7	Unmated probes placed in the gallery	QMUL rank-1 reported as 0.53%	Protocol roles enforced; locked by an invariant test
8	Genuine pairs built by sort order	TinyFace rank-1 inflated to 37.43%; 39.4% of pairs were gallery-to-gallery near-duplicate track frames	Official gallery-match by probe only
9a/9b	Upstream build path and keypoint units in the ViT wrapper	LFW 56.70%	Absolute paths, restored working directory, normalised keypoints
10	Canonical keypoints on unaligned crops	TinyFace 15.19% — below the incumbent	Published alignment path made the default and contract-tested

#	Defect	Observed effect	Fix
11	Result artifacts written without a model key	Each backbone's run destroyed the previous one's artifact	Model key in the filename

13.1 Withdrawn figures

Withdrawn	Replacement	Cause
TinyFace rank-1 37.43%	32.93%	Defect 8
TinyFace TPIR at 1% FPIR 9.60%	Unmeasurable — withdrawn	TinyFace has no unmated probes, so the quantity measured nothing
QMUL rank-1 0.53%	2.68%	Defect 7
All capacity figures on unlabelled protocol pools	None — refused	Saturation and contamination could not be excluded from pools without identity labels
IJB-C 86.83% / IJB-B 43.43% at 1e-6	86.73% / 43.13%	Quantile interpolation replaced by the exact order statistic

A defect in a test, recorded because it was ours

An audit of an earlier draft found that the first regression test for defect 8 was not doing what its docstring claimed: its assertions checked a row count and an array dimension, and it imported the wrong loader module entirely. It was replaced with one asserting exact row counts and the recorded genuine-pair count on the correct module. **A test that looks like a lock and is not is worse than no test**, because it converts an unchecked assumption into a checked one in the reader's mind without doing any checking.

14. Hardware envelope, and what it excluded

All work was carried out on a single NVIDIA RTX A3000 Laptop GPU with 6 GB of memory. That ceiling is not incidental — it decided several design choices, and each exclusion is recorded so that a reviewer can distinguish a limitation of the method from a limitation of the machine.

Excluded	Requirement	Consequence
SUPIR (SDXL-based restorer)	12–24 GB	Best perceptual quality in the field and not available to us. Revisit only on a 16 GB or larger card
Full fine-tuning of the ViT backbone	Optimiser state for 115M parameters at a usable batch size	Replaced with rank-8 LoRA at 1.2% of parameters — which is also the better scientific choice, since the prior full fine-tune failed partly through catastrophic forgetting of calibration
Larger backbones and longer pretraining	Memory and time	Rejected on evidence as well as cost: the clean-pack table shows the ViT already matches the R50 there, so scale is not the constraint — condition is
Training a generative renderer	Multi-year cost and memory far beyond this envelope	Architecturally excluded from the evidential path regardless

A second, independent constraint is data. Forensic surveillance corpora are small and licence-restricted. SCface, for example, was used strictly as an *evaluation* corpus for the pre-registered comparison in §9.3 and was never used as training data. Both constraints bound the ceiling of the results reported here.

15. Reproducibility

- **Every figure names its artifact.** Results are written to model-keyed JSON files, and each table in the measurement record names the file that produced it.
- **The harness is deterministic.** Re-running the R50 baselines after the artifact-naming fix reproduced their recorded values exactly — TinyFace rank-1 32.93% and Cllr 0.6662, QMUL rank-1 2.68% and TPIR 0.00%.
- **Two independent code paths agree.** The IJB-B re-run reproduced every operating point exactly and matched figures computed independently by a separate feasibility script from the same cached embeddings.
- **A regression gate protects the numbers.** Any model, threshold or fusion change triggers the benchmark suite across five datasets and fails loudly on drift.
- **Contract tests protect the assumptions.** Seven tests lock the ViT integration, including an assertion that alignment remains the default path; further tests lock the protocol invariants behind defects 7 and 8.

16. Limitations and roadmap

#	Limitation
1	**No independent validation.** Every figure was produced by the same team and tooling that built the system. Nothing internal can close this.
2	**The published-ArcFace comparison band is uncited** and must be pinned to a specific paper and table before the harness-validation claim is relied upon externally.
3	**Open-set identification on real cross-camera surveillance remains near zero** — 1.38% true-positive identification at 1% false-positive identification rate after fine-tuning, against 0.00% before it.
4	**Condition leakage failed its registered criterion** in the LoRA run and requires a second experiment with the adversarial term trained.

#	Limitation
5	**Demographic differentials persist** and must be re-measured on the deploying population.
6	**CFP-FP absolute values** were measured on an outlier protocol pack and understate all configurations by roughly 1.5 points.
7	**Capacity estimates on TinyFace are a floor,** not a resolved value, because censoring reached 13.1% after the backbone swap.
8	**Open-set generalisation rests on a single corpus.** TinyFace structurally cannot measure it, so QMUL is the only open-set evidence. A second open-set corpus remains an outstanding acquisition.
9	**Full-stack latency is unmeasured.** The reported figures are engine-level.

16.1 Next steps, in order

Phase	Work	Gated on
8	NIST readiness — CPU accuracy and latency sweep, submission-path parity harness, validation package	Nothing; it blocks the rest
9	Ship the measured wins — quality-routed selection for 1:1, exact index in production, score normalisation and WCCN	Phase 8 findings
10	Degraded imagery — second LoRA run with the leakage term, further real low-resolution data, demographic differentials as a first-class result	Phase 9
11	Submit — FRTE 1:1, FRTE 1:N, and the quality-assessment track, which is the closest existing asset to a track-leading position	Phase 10
13	Publish — the capacity result: single-image open-set identification on cross-camera surveillance is information-limited rather than architecture-limited, with the bit budget that shows it	Phase 11

Closing statement

The recognition model is not this programme's contribution and is not presented as one. The contribution is a measurement framework that made a falsifiable prediction — that a better backbone would improve ranking and leave open-set identification unchanged — and was confirmed by the experiment that could have refuted it. The accompanying result, that single-image open-set identification on cross-camera surveillance is limited by available identity information rather than by model architecture, is more valuable stated honestly with its bit budget than any marginal accuracy claim would be.